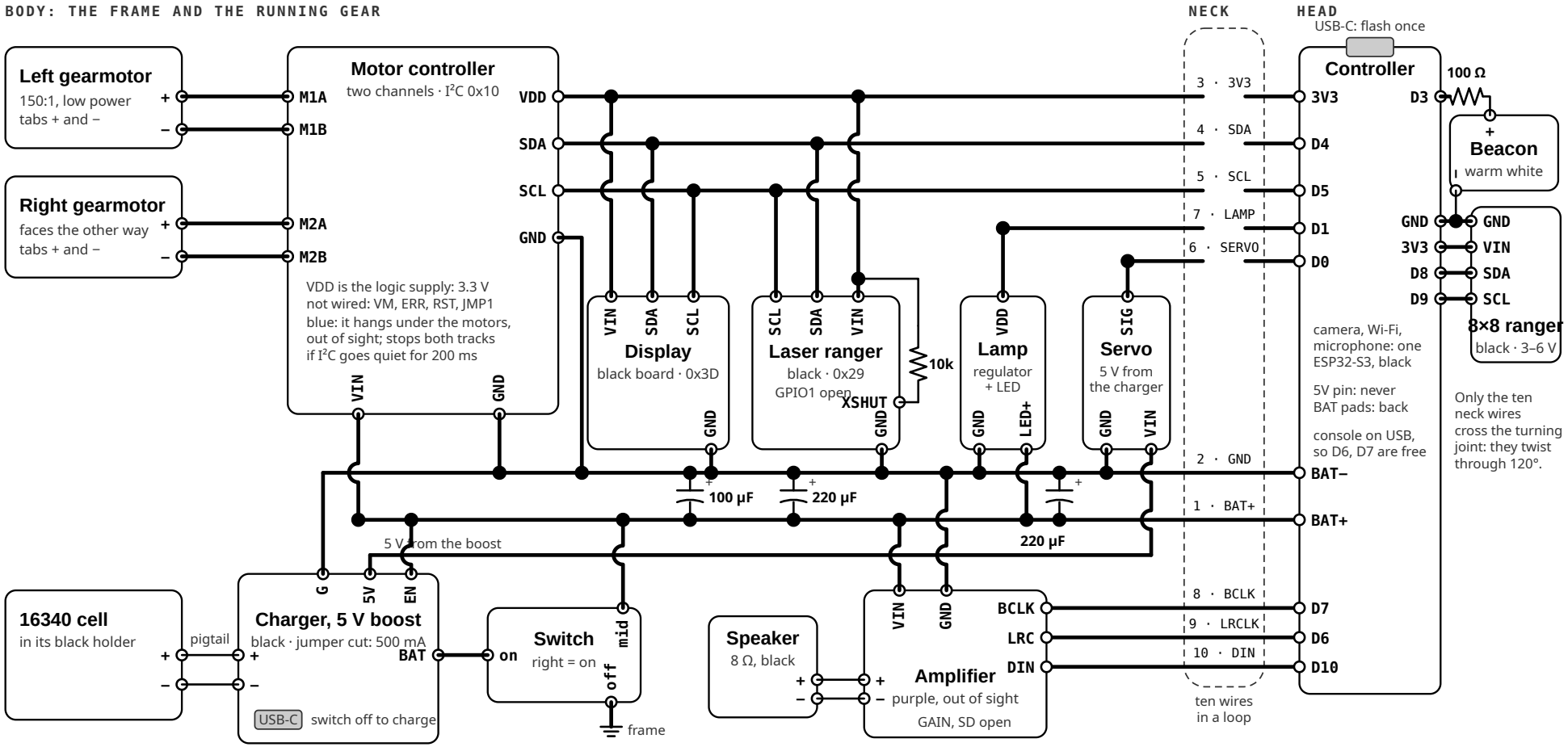
Mars rover: every connection on one sheet

A dot is a joint. A line that hops over another does not touch it. Thick = wire you solder (26 AWG for BAT+, 30 AWG for the rest). Thin = a lead that comes with the part. Pins are grouped by job, not by position on the board.



The budget, from the cell (mA)

parked, watching	158 mA	Wi-Fi on, camera idle, display, ranger · 5.4 h
exploring	446 mA	camera streaming, the 8×8 ranger on, both tracks, the head sweeping · 1.9 h
at night	621 mA	exploring + the flashlight at its 50 % cap · 1.4 h
talking	278 mA	stopped, listening and speaking at a room's level · 3.1 h
never all at once	2002 mA	every stall and the lamp full; ramps and caps keep peaks near 1 A

3.3 V rail at its worst: 572 mA of the regulator's 700.
Charge: USB-C at the charger, port up, switch off, 500 mA, about 2.2 h.
Never: both USB-C ports at once · the 5V pin · the cell soldered.
Flash with the switch on, so the board's own charger sees the cell.
In the rails: BAT+, 3V3, SDA, SCL left; BCLK, LRCLK, DIN, 5V right.

Rev K for approval, 2026-11-10: a black charger with its own 5 V boost, the voice on three more neck wires, the switch grounding the supply when off. 15 crossings, each a hop. Sources: Pololu, Adafruit, Seeed, Same Sky, ST, Analog Devices.